Keep direction attached to every arrow

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Resultant / ES: resultante / PT: resultante

The overall arrow from the first start to the final finish.

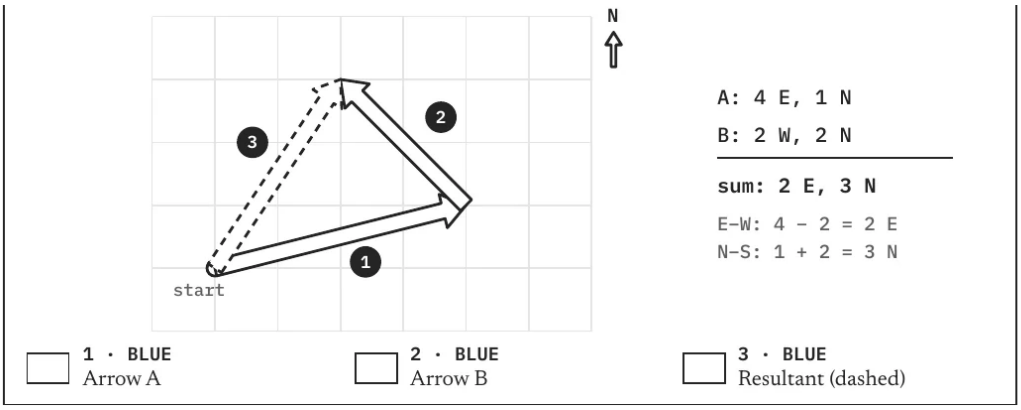
Also on your note page: Vector, Tip-to-tail, Component.

B READ THE RELATIONSHIP

A resultant is one arrow with the same overall effect as the arrows you add.

Add signed components along each axis

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 20 N pull points 30° above the +x axis. Split it into its x and y parts.

Given F = 20 N θ = 30° from +x

Rule F x = F cos θ F y = F sin θ

Work 20 cos 30° 20 sin 30°

Answer 17.3 N, 10.0 N parts of one force, not two new forces

E SAVE FOR THE FINAL MEETING

XJ's motion has parts in more than one direction. Adding them tip to tail gives its true path; splitting them shows the part that actually closes on Earth.

Read the physics: Resultant

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

A vector has a size and a direction. Draw the next arrow with its tail at the first arrow's tip. Do not turn the arrow when you move it.

02 HOW IT WORKS

Draw the resultant from the original start to the final tip. To calculate, add horizontal parts together. Add vertical parts separately. The two totals make a right triangle.

03 WHAT TO CHECK

Use the sides to find the length and the signs to choose the direction. State the axis you measured the angle from.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 20 N pull points 30° above the +x axis. Split it into its x and y parts. 17.3 N, 10.0 N parts of one force, not two new forces.

05 CONNECT TO XJ

XJ's motion has parts in more than one direction. Adding them tip to tail gives its true path; splitting them shows the part that actually closes on Earth.

SAY IT BEFORE YOU WRITE

Where does the resultant start and end? The two parts are ___ and ___. Together they point ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Vector

ES: vector / PT: vetor

Meaning in my words	How can you tell?
How would you explain vector to a friend? In my words, this means ___.	What must you tell a partner so they can draw the same displacement arrow? I can tell because ___.
Example or sketch	**Non-example and why**
Using 1 cm for 1 m, how would you draw 3 m west? My example shows ___.	Someone labels a trip “3 m” and stops. What information would make it a vector? I would change ___ because ___.

Tip-to-tail

ES: punta con cola / PT: ponta com cauda

Meaning in my words	How can you tell?
How would you explain tip-to-tail to a friend? In my words, this means ___.	Where do you put the tail of the north arrow after drawing the east arrow? I can tell because ___.
Example or sketch	**Non-example and why**
Draw 3 m east followed by 2 m north. Where is the final tip? My example shows ___.	Can you turn the north arrow to make the two arrows fit on the page? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Component

ES: componente / PT: componente

Meaning in my words	How can you tell?
How would you explain component to a friend? In my words, this means ___.	If east and north are positive, which component is negative for a southeast trip? I can tell because ___.
Example or sketch	**Non-example and why**
What two arrows could you use to build a displacement of 3 m east and 4 m north? My example shows ___.	A classmate counts the diagonal and both parts as three separate trips. What is counted twice? I would change ___ because ___.

Resultant

ES: resultante / PT: resultante

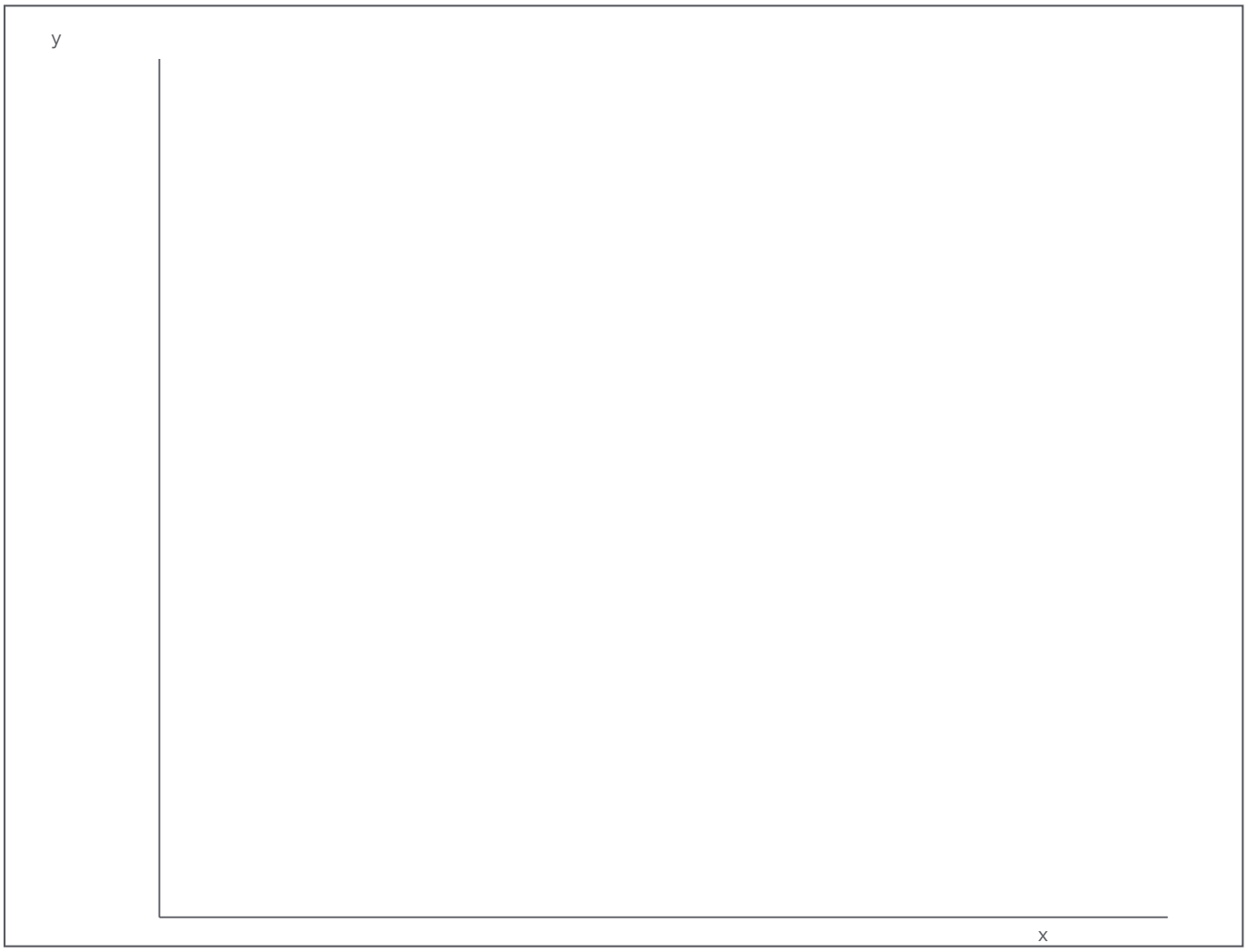
Meaning in my words	How can you tell?
How would you explain resultant to a friend? In my words, this means ___.	Where does the single arrow start and end on a tip-to-tail drawing? I can tell because ___.
Example or sketch	**Non-example and why**
Walk 3 m east, then 4 m north. What single arrow joins your start and finish? My example shows ___.	Someone labels the resultant 7 m. Which length did they add up? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a right-angle journey; label x, y, and the resultant with its direction.

Labels you can use: horizontal · vertical · resultant.



Tell a partner

The two parts are ___ and ___. Together they point ___.

Send the whole movement

06

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Combine two displacement components into one arrow.

CHOOSE ONE JOB TOGETHER

NAVIGATION ENGINEER ☐

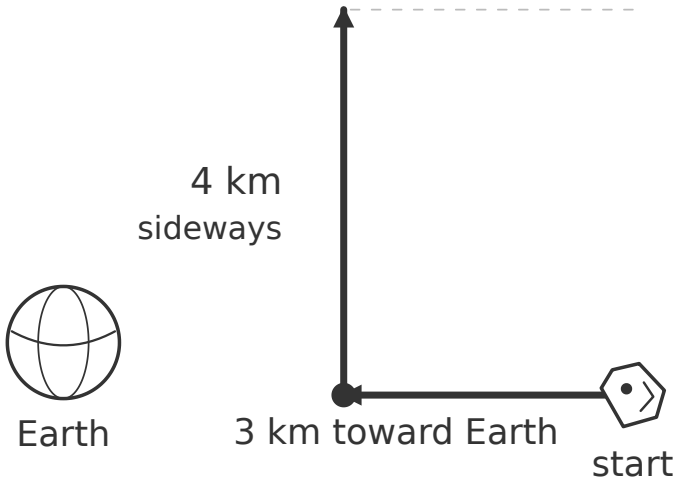
Draw one arrow for the whole displacement. Give its length and direction, and label both component legs.

MAP PRESENTER ☐

Use the map to report the whole movement. Explain why '**7 km** closer to Earth' is the wrong caption.

Word help: component / sideways / resultant

A tracker sees a **3 km** displacement toward Earth and **4 km** sideways. The second telescope needs the whole movement.



01 OUR FIRST CALL

What full movement should the second telescope track? Use your map.

The whole displacement is ___ toward ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What part can the updated tracker measure? What part will it miss?

Now we would say ___ because ___.

Switch speakers. Does your arrow show both parts of the movement?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks